



Fig. 6: Launchers in Aegis destroyer (top) and Aegis cruiser (bottom)

not fixed in any mast but embedded on the superstructure of the vessel itself.

Using this technique, many individual beams can be formed depending upon the requirement for various functions, with each beam performing a specific function like scanning, tracking or providing data-links. A single radar antenna covers a 90° sector in the azimuth and from the horizon in the front to the zenith overhead in elevation.

There are four antennae for this radar to cover a total of 360°. In this arrangement, a hemisphere of RF energy is maintained around the ship up to a range of around 185km (100-nautical-miles). The radar's peak power is 4MW to 6MW and average power of 58kW in the S band.

Next in the line are the missiles for engaging the target.

SM-2 SAMs. The surface-to-air missile (SAM) used in Aegis vessels is called Standard Missile-2 (SM-2). Its range is quite long and current versions are capable of engaging targets even up to 200km. The guidance scheme of SM-2 is a semi-active radar-guidance scheme.

Generally, due to the long range involved, majority of long-range SAMs have semi-active radar-guidance schemes. This means that these do not have radar transmitters but only radar receivers. These can only receive radar echoes but not chirp out radar signals like usual radars. These depend on other radars for guiding these to intercept the target.

Due to the longer engagement range, more radar signal strength is required to look for the target. For more signal strength, the missile must have ultra-strong batteries to pump enough amperes into the radar. But ultra-strong batteries are heavy and consume more space in the missile. This extra weight introduces restrictions in the missile's manoeuvrability.

Due to these reasons, the missile only has the radar receiver section as the target seeker. As reception does not involve huge power requirements, the receiver fits snugly into the requirement. The ship, which launches the missile, provides radar energy through an illumination-/fire-control radar. This radar does not chirp out pulses but continuously focuses its high-energy radar signals on the incoming target. Reflected energy is picked up by the missile seeker.

The missile's guidance system understands the position of the target due to the directional property of this received radar energy. Then, the guidance system aligns the missile for successful interception and intercepts the target to destroy it.

In Aegis cruisers and destroyers, SM-2 missiles are stored in a vertical launching system (VLS), which can hold 122 missiles in total and those present in destroyers can hold 96 missiles.

VLS is a generalised launcher system and can launch many types of missiles. Though depending on mission requirement, various types of missiles could be loaded to this VLS.

MECO®

SINCE 1962

LOW COST LCD / LED MODULES



LC035



LC135

Specifications

3½ Digits, 1999 Counts (max.) LCD	3½ Digits, 1999 Counts (max.) LED
LC035	LC135
0 - 200mV DC	0 - 200mV DC
0 - 2, 20, 200V DC	0 - 200V DC
0 - 200µA DC	0 - 2, 20, 200mA DC
Auxiliary Power Supply	
9VDC $\pm 10\%$	5VDC $\pm 10\%$

Applications

- Ammeter
- Voltmeter
- Thermometer
- Power Supply
- PH Meter
- DB Meter
- Watt Meter
- LUX Meter
- LCR Meter
- Capacitance Meter
- Other Industrial and Domestic uses.

An ISO 9001:2008 Certified Company

MECO INSTRUMENTS PRIVATE LTD.

Plot No. EL-1, MIDC Electronic Zone,
TTC Industrial Area, Mahape,

Navi Mumbai - 400710 (INDIA)

Tel : 0091-22-2767 3300 (Board), 2767 3311-16 (Sales)

Fax : 0091-22-27673310, 27673330

Email : sales@meconinst.com

Web : www.meconinst.com